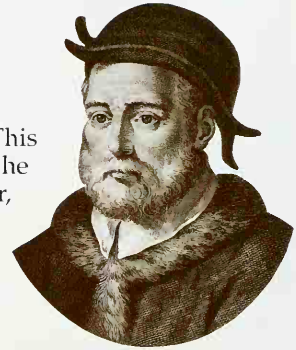
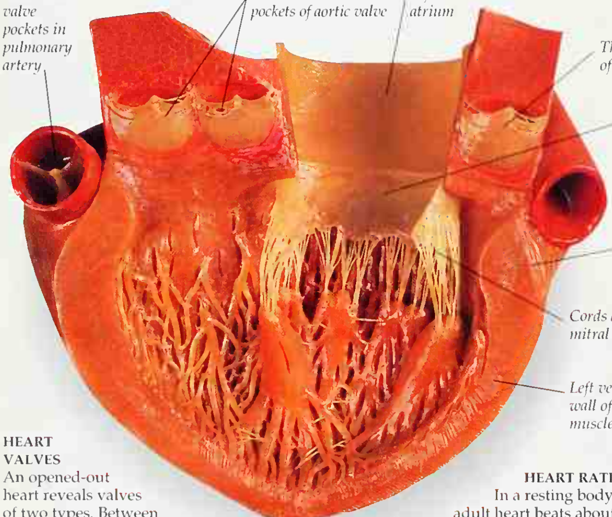
The heart

SACRED HEART
In Christian symbolism, the heart represents the source of goodness and love. Jesus said "Lay up for yourselves treasures in heaven .... For where your treasure is, there will your heart also be."

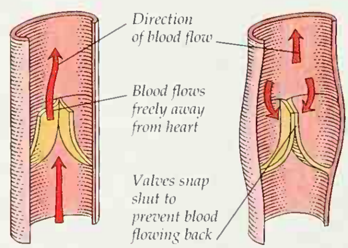
THE HEART IS A STRAIGHTFORWARD, extraordinarily reliable, muscular pump. Its anatomy is relatively simple, yet few body parts have caused as much confusion in their description. In the 4th century BC Aristotle wrote that the heart had three chambers. This is true in creatures such as frogs and lizards, which he had probably examined. The human heart, however, like that of any mammal, has four chambers. There are two on each side: an upper, thin-walled atrium and a much larger thick-walled ventricle below. Aristotle also stated that the heart was the seat of intelligence, and this notion persisted in various forms for centuries. Our language today has many references to the heart as the center of desire, jealousy, loyalty, love, courage, and other emotions, although the brain is the true site. Galen and many people after him said that the heart's central dividing wall, the muscular septum, had tiny holes or pores. These allowed blood to seep from one side of the heart to the other. Andreas Vesalius tried to find them by poking hairy bristles at the septum, but he failed – because there are no such pores. Only with William Harvey's work (p. 30) did the heart's real role as a pump become clear.



THE RIGHT CONNECTIONS
Italian anatomist and botanist Andrea Cesalpino (1519-1603) produced a remarkably accurate description of how the heart is plumbed into the main blood vessels and connected to the lungs – more than 20 years before Harvey's explanation in *De Motu Cordis*. However, in Cesalpino's last book, he incorrectly stated that the blood flowed out of the heart along all vessels, the veins as well as the arteries.



HEART VALVES
An opened-out heart reveals valves of two types. Between each atrium above ("entrance chamber") and its ventricle below is a large valve anchored by cords. This is the mitral valve on the body's left side; on the right is the tricuspid valve. The valves prevent a backflow of blood from ventricle to atrium. The cords stop the valves flipping inside out, like an umbrella in high wind. As the ventricles squeeze, blood flows into the arteries (p. 31) through semilunar valves.



VALVES AT WORK
The semilunar valves at the two exits from the heart have pocket-shaped flaps of tissue fixed strongly into the arterial wall. Blood can push its way out, flattening the flexible pockets against the wall. When the heart pauses before its next beat, the highly pressurized blood in the artery tries to flow back. This opens out the pockets, which balloon together to make a bloodproof seal.

HEART RATE AND PULSE
In a resting body, the average adult heart beats about 60-80 times, pumping around 1.3 gal (6 liters) of blood every minute. Each beat creates a pressure surge through the artery network. The surge, which can be felt in the radial artery in the wrist, is called the pulse. During exercise, the muscles demand more oxygen and nutrients. The heart beats faster and harder, as much as 150 times a minute, to circulate up to 8.8 gal (40 liters) of blood.